

Evaluation of techniques for manufacturing process analysis

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Abstract In the last 20 years, several methodologies, models and tools have been developed for the analysis and optimisation of manufacturing systems in order to propose general improvements. Many of these techniques make extensive use of data modelling, simulation, decision-making support, expert systems and reference models. This paper presents the first outcome of a piece of research work to integrate manufacturing process analysis into an integrated modelling framework covering all aspects related to the shop-floor as it really is. The main methodologies and software tools have been identified and evaluated and the results tested on industrial examples. As a result of this evaluation it has been possible to identify the inefficiencies of the techniques. These problems are connected with integrating the different types of data to be analysed—such as quality, time, costs, resource capacity, productivity, flexibility or improvements—into a single analysis environment. The inefficiencies detected enable us to present a general framework for making better use of modelling techniques for manufacturing process analysis.

Keywords IDEF · BPM · Process modelling · Simulation · Manufacturing analysis · KPI's

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Introduction

In today's highly competitive global industry, the demand for high quality products manufactured at low costs with shorter cycle times has forced manufacturing industries to consider various new product designs, manufacturing, information systems and management strategies. In this decision process, the application of system engineering methods and tools to the modelling, analysis and optimisation of manufacturing systems is the best way to achieve the goals proposed. Decision support systems, software process modelling, expert systems, business process reengineering, simulation software, ABC cost manufacturing models and manufacturing databases are some of the multitude of methodologies and tools that allow to analyse a manufacturing system and support the decision-making process.

This paper evaluates the use of these methods and tools in order to propose general improvements in a manufacturing system and evaluate the implementation of new techniques. The review assesses those initiatives attempting to integrate broader ranges of factors. The results have enabled us to identify unresolved problems.

The paper begins by providing a brief overview of the manufacturing areas involved for the analysis and improving of manufacturing systems. Following this is an in-depth review of techniques that can be used to perform a manufacturing analysis. The third section presents the final conclusions of evaluation, showing the disadvantages of the techniques discussed in the previous section. The paper concludes by suggesting a general framework focused on making better use of modern modelling techniques and system engineering tools.

General environment for analysing and improving manufacturing systems

Manufacturing analysis for continuous improvement is a technical area with high significance due to the increase of quality and flexibility requirements for final customers. There are many applications cases where manufacturing analysis is performed in order to take decisions:

- Diagnosis of wastes (capacity, resources or cost).
- Diagnosis of a disorder (material, information or control flow).
- Restructuring a manufacturing process to improve its performance.
- Business process Reengineering (BPR).
- Implementation of Enterprise Resource Planning (ERP), Manufacturing Execution Systems (MES) and Product Data Management (PDM).
- Implementation of JIT, TQM, TPM, QFD strategies.
- Implementation of ABC cost accounting method.
- Tuning the organisation structure to face business change.
- Large scale systems integration.
- Alignment or conformity to norms (ISO 9000, ISO 14000).
- Management decision (activity externalisation/internationalisation).

Figure 1 shows the phases of a full manufacturing analysis and a general view of the areas involved. In a general way, a manufacturing process analysis starts with an initial phase, in which the system is analysed using traditional engineering methods and systems engineering techniques. The results of the first phase allow the strong and weak points to be identified, as well as the activities that must be improved to get the best performance. In the second phase, decisions are made in order to select those optimal improvement techniques for the specific situation of the production system. The final phase consists in the measurement of results in order to reconsider future actions in the cyclical process of continuous improvement towards excellence. This last process is carried out with the measurement of key performance indicators (KPIs) and the application of the latest activity-based costing/management (ABC/ABM) techniques. Usually, decisions on improvement in companies are taken using summary data, ratios and indexes that are generally calculated using methods developed for each particular case.

The added value of these actions in modern industry suggests developing new standards methods and tools that are able to integrate the three above-mentioned

phases into a single environment. Most of the research papers related to general manufacturing analysis are conceptual works or specific case studies (Felix & Bing, 2001; Guimaraes, 1997; Gunasekaran & Nath, 1997). In all cases, the goal is to obtain performance data like lead time, productivity, cost, flexibility or product quality in order to decide the best solution for a specific system. However, there are inefficiencies, which is the reason why many companies do not use these methods in practice. The next section presents an in-depth review of the main techniques that can be used in a general analysis of a manufacturing system.

Techniques for manufacturing analysis

This section shows a critical review of the state of the art of the techniques described in Fig. 1. The review has been carried out taking account of the practical contribution of each technique to the general process of analysing a manufacturing system. The diversity of existing techniques called for grouping the techniques examined into four major groups, in order to make an individual analysis of each of them. The groups identified were: diagnostic reference models, information modelling, dynamic simulation and integrated modelling methods.

Diagnostic reference models

Usually, the real implementation of new production management techniques (JIT, 5S, SMED, TPM, Lean Manufacturing, Concurrent Engineering, ERP, MES) is based on methodologies that permit an analysis of their state with respect to reference models.

A first group of these methodologies is made up of those that are based on questionnaires, with questions designed to identify the current state of the production system. The results are compared with reference tables that permit a calculation of the present theoretical performance of the production system. Aquilano and Chase (1991) drew up this type of questionnaire, which, over time, has been adapted and employed by many industrial and consultancy companies. These questionnaires group the principal aspects that must be studied in a production system into different categories or levels of improvement. Normally, they analyse the product structure, the manufacturing flows, the co-ordination of the plant with other functional areas, the facilities, production planning, programming systems and the workforce. In most cases, these methods are accompanied by IT tools which make it possible to automate their application. This is the case of Vollmann, Dixon, and Nannu

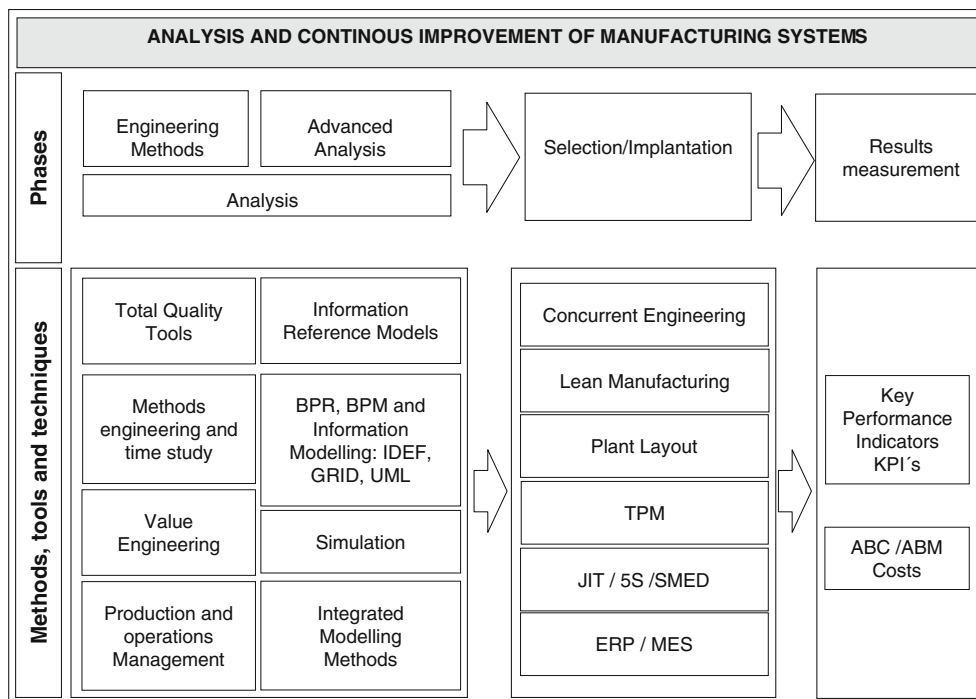


Fig. 1 Methods, tools and techniques in manufacturing analysis

(1989), who created a diagnostic tool and a method for analysing the measurement requirements of a production plant. This tool comprises a series of questions designed to measure the chief indicators and to establish the deviations from proposed standards. Logically, the results depend on the correct configuration of the standard model. In the same way, the diagnostic method proposed by Jackson (1996) is very interesting as it scores and evaluates the answers to specific lists of questions. Answers are linked with actions and goals in order to transform the system towards lean manufacturing.

A second group of methodologies is formed by those that use procedures for measuring the principal system factors such as productivity, costs or flexibility and establishing how to match a reference model that is considered optimal. Among the first initiatives of this type, Seidel (1998) deserves a mention. He developed a methodology oriented towards the evaluation and improvement of productivity in job-shop companies, starting from a model in which activities in eight classes are categorised: purchases, engineering, plant distribution, planning, manufacturing, programming, customers and design. The author represents the relationship between the different classes by means of one matrix of productivity and another of problem identification factors. Both matrices allow to quantify the possibilities for improvement of each of the basic system activities, by calculating a global productivity matrix that indicates those spe-

cific areas that must be targeted for improvement, given the lack of performance in their output. Wiendahl and Ullmann (1993) also developed a specific method for measuring plant productivity, based on a model that analyses production flows by means of a “throughput diagram”, which registers the inventories for each product generated throughout the process activities. The method gave rise to a monitoring tool that enables data to be compiled on the performance of the manufacturing system outputs and points the way towards the ideal course of action for continuous improvement. Rozenfeld, Rentes, and König (1994) proposed a structure of information metadata, based on the configuration of five models: business, operations, data, resources and organisation. From this initial structure, a modelling methodology was developed, aimed at reengineering processes in CIM contexts. Its main contribution lies in proposing an information structure that enables a better understanding of the company processes related to manufacturing. Miltenburg (1996) developed a complete framework for formulating manufacturing strategy, which establishes an analytical method that identifies the state of a manufacturing system, within the evolution towards industrial excellence. When it comes to being able to make an assessment of the system, the method provides a clear vision of what the key indicators are, and identifies the series of indicators related to system outputs. Identifying and acting on these indicators (manufacturing

levers) paves the way to proposals for the improvement technique that produces the best results.

An interesting concept in relation to the improvement of manufacturing systems through diagnostic models has been the inclusion of the “value added” concept in these reference models. In this sense, Trischler (1996) developed a methodology that allows systems to be analysed through the identification of activities that do not contribute added value to a process. The methodology establishes the phases through which a project for improving processes should develop, and its main contribution is a reference model based on a “dictionary of activities” that do not contribute value.

Nowadays, diagnostic reference models have integrated new techniques of systems engineering. This is the case of Heredia (1999) who proposed a reference model for the integrated management of processes, comprising a model of organisation by processes, a model of indicators and a data model. Its primary goal is to serve as a guide for implementing a system of strategic indicators for the integrated management of a company, based on the principles of total quality. The same author participated in the European project ESPRIT TQM-Til, which has the goal of developing a Web-Tool that will permit the exploitation of indicators from corporate OLAP databases.

Information modelling

There can be no doubt that information modelling is the most important technique used for manufacturing analysis. During the last two decades several process modelling methodologies used for knowledge capturing have been developed and represent the structure, behaviour, components, resources and operations of a manufacturing system to understand, reengineer, evaluate and optimise performances. Figure 2 shows the history of modelling techniques.

Based on these techniques, a high number of process modelling tools have been developed, e.g., ARIS (IDS Scheer), Matis (NCR), FirstStep (Interfacing Technologies), MM/IEM (IPK), CimTool (RGCP), Grai-Tools (Graisoft), Scitor (Sciforma), Structware (Temas) or All-Fusion Modeller (Computer Associates). The links at <http://www.is.twi.tudelft.nl/~hommes/toolsub.html> and <http://www.icaen.uiowa.edu/~coneng> provide an extensive overview of process modelling tools.

Of all the methodologies shown above, the IDEF family and mainly IDEF0 (NIST, 1993) and IDEF3 (Mayer & Menzel, 1995) are the simplest and most widespread in an industrial context for a high variety of purposes. The IDEF0 definition of a system is a set of activities that takes certain inputs and, using some mechanism, subject

to certain controls, transforms those inputs into outputs. Such objects can be used to model relationships among various activities. The flexibility of the IDEF resides in its capacity to allow the analysis of complex systems, where there is a need to study multiple levels of detail of activities in a way the analyst can understand the system. In the literature review, a lot of end applications showing the high adaptability of the method can be found. Their end applications are highly diverse, covering:

- Modelling for how to implement design techniques for manufacturing (Colquhoun, Gamble, & Braines, 1989; Guiachetti, 1999) or concurrent engineering (Barreiro, Labarga, Vizan, & Rios, 2003; Howard & Lewis, 2003).
- Study of how to implement ERP systems (Johnny, Ip, & Lee, 1998; Kwon & Lee, 2001).
- Capturing information to be used in simulation (Lingineni, Caraway, and Benhamin (1995), Thompson (1995), Perera and Liyanage (2000)).
- Evaluation of computer technologies to be introduced into small and medium size manufacturing enterprises (Sarkis & Liles, 1995; Tatsiopoulus, Panayiotou, & Ponis, 2002).
- Representing reliability-centred maintenance (RCM) business process to decide the maintenance strategies (Gabbar, Yamashita, Suzuki, & Shimada, 2003) or supporting a fault tree based methodology for reliability evaluation and risk with IDEF3 (Kusiak & Zakarian, 1996a, b).
- Developing a database system to optimise manufacturing processes during design (Howard & Lewis, 2003).

All these applications show the significance of IDEF0 and IDEF3 modelling for understanding and analysing a manufacturing system. However, IDEF is a descriptive method based on the graphical and text description of functions, information and data that is often only of use as an aid to documenting and validating processes. It is static and qualitative, which is a drawback to the analysis of processes. It is difficult to compare the static model with the physical process since times are not represented within the model. Zakarian and Kusiak (2000) affirm that the activities in an IDEF model are at a relatively high level of abstraction making it difficult to associate exact quantitative data for the process variable of interest. This is one among the reasons why many developments based on IDEF oriented to proposing general improvements have several limitations.

Several researchers have used IDEF as a basic technique to tackle manufacturing analysis. Bilberg and

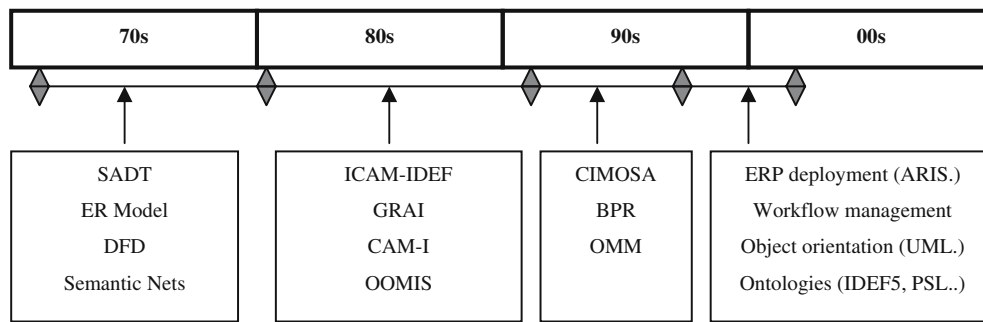


Fig. 2 Enterprise modelling history and background

Alting (1994) used IDEF0 modelling as an analysis method for putting forward conceptual solutions for improvements in a production system. Nevertheless, although they proposed the use of simulation tools to evaluate these solutions, they did not address the issue of their integration, thus reducing their applicability for obtaining an activity model that may facilitate subsequent diagnosis. The true potential of IDEF0 arises when mechanisms for incorporating quantitative information are created. In this sense, Mayer and White (1991) proposed the use of IDEF0 models for analysing the activities of an improvement project and the later identification of cost drivers by means of Analysis Based Costing (ABC) techniques. ABC is a cost accounting method that identifies activities performed, establishes the cost of each activity, and traces the cost of the activities to the product. The method developed by Mayer and White is carried out using the information from the IDEF model, and exporting it to a project management program where, by adjusting the time and cost factors, the analyst can design the optimum planning for the project. Developments already available in this line of work include the Smartcost tool, which from an IDEF0 model feeds data into an Excel spreadsheet template. The template permits the application of cost techniques, based on activities, in the detailed study of manufacturing costs. Silver (1997) defends this kind of integration of IDEF0 modelling with workflow tools. His developments have given rise to the Workflow Analyser tool that supports the complete set of rules and conventions employed in modelling IDEF0 processes, and its integration into a method that permits the definition of the dynamic behaviour of activities (duration of activities and consumption of resources). This is a very interesting aspect, since it transforms a static diagram of an IDEF0 process into a dynamic simulation model that can be used for analysing bottlenecks and the application of activity-based cost systems.

The techniques shown above resolve specific problems in a manufacturing environment. However, a

general manufacturing improvement analysis needs the formulation of a well-defined general manufacturing strategy. For this purpose, other modelling methodologies such as GRAI-Grid (Doumeingts, 1985) help to define the most important decision selecting processes and information among the different functional areas of the enterprise. The main outcome of this model is the identification of improvement areas which are in line with the general strategic orientation. The decision centres identified in the GRAI Grid are decomposed into detailed IDEF0 diagrams. The final model may be improved by using IDEF3 to get a more analytical and behavioural representation able to support process simulation. This methodology has continued to be developed with successive adaptations and applications (Girard & Doumeingts, 2004).

There are other theoretical approaches to overcome the limitations of IDEF using techniques such as Fuzzy Logic or Petri Nets. Bosil, Giaglis, and Hlupic (2000) have evaluated the suitability of IDEF and IDEF3 in conjunction with Petri Nets for modelling processes. Ma, Zhang, and Ma (2002) have developed a formal framework to provide extensions to IDEF1X to represent fuzzy information. In both cases, the real application to end-users is difficult due to the complexity of the techniques.

Currently, tendencies in manufacturing modelling are focused towards the use of UML language as a complementary modelling method. In its origin, UML is a modelling language that can be used to generate computer-executable models that encode key aspects of software engineering projects, but it can also be used for process modelling. Kim, Weston, Hodgson, and Lee (2003) provide a comprehensive review of IDEF techniques and UML. They identified similarities between IDEF objects and UML objects. They have observed that the combined development and reuse of IDEF and UML models has the potential of linking the description of the manufacturing system and the behaviour of the enterprise objects such as suppliers, products, machines,

people, teams, etc. UML is used too by Habchi and Berchet (2003) to model the control process of production into simulation models. It is clear that this technique provides a new opening for improving and exploiting the data stored in a manufacturing model.

Dynamic simulation

The simulation technique has been used since the 1960s as a tool for investigating the underlying behaviour of many types of manufacturing systems. Over the last two decades, there has been a dramatic increase in the use of simulation to design and optimise manufacturing and warehousing systems. The reason can be found in the facility for evaluating the movement of parts through the machines and workstations, and examining the conflicting demand for limited resources.

Simulation is performed through the use of one of the many commercially available tools (Table 1). Many of them have served as the basis for numerous works that analyse and compare the applicability of these systems within the manufacturing sector. The results of the works of Tillal and Ray (2001) and Hlupic and Paul (1999) indicate that the applicability of simulation is greater when it comes to solving specific problems in concrete sectors. We can refer to several topics: production planning (Grabau & Mauer, 1997), bottlenecks (Neely, 1992), scheduling (Williams & Naraya, 1997), study of wastes (De Smet & Gelders, 1997), Just-In-Time system design (Soon & Souza, 1997), reengineering modelling (Irani, Hlupic, Baldwin, & Love, 2000).

Simulation software provides user interfaces that allow significant reductions in programming time, but the need for well-defined simulation model specifications is necessary. In this sense, the complementary use of modelling techniques enhances the quality of simulation models and reduces the time needed to configure them. This is the main reason for many researchers to focus their developments on obtaining a direct connection between modelling tools and process simulators. The first initiatives in this area stemmed from Lingineni et al. (1995) and Thompson (1995), who developed tools for capturing information from IDEF3 models and integrating it into the Witness simulation package. Rojas and Martínez (1998) developed a prototype that supports the capture, description and analysis of activity process models for the automatic generation of code for computer simulation tools. As an activity modeller, the prototype uses the Role Activity Diagram (RAD) technique (Ould, 1995). In addition to the description of RAD activities, it includes information on the objectives and decisions associated with the development of the model. The prototype is based on the development

of an interface between RAD models and the Witness simulation tool. It establishes a map of correspondences between the elements of the model (activities, relations, restrictions, decisions...) and the elements of the simulator (machines, buffers, products). Nevertheless, the RAD model does not contain information on the attributes related to the dynamic properties of the activities (cycle times, times of arrival, amounts...), with the result that it is necessary to include this information independently, through a specific user interface. In this way, the basic information sent from the model to the simulator is enhanced, enabling it to be transformed into Witness input commands. Perere and Liyanage (2000) developed a fast methodology for compiling data that can be used in a manufacturing systems simulator. They propose a library of reference models for processes, on the basis of a set of IDEF0 diagrams and a data reference model (IDEF1X). As a result, databases that accelerate the task of information gathering are obtained. Jeong (2000) also proposes the use of IDEF0 to model the information necessary to feed simulation-based production programming systems (OSBSS). The author uses IDEF1X and IDEF3 methods to generate the simulation model within an advanced architecture that makes use of a database of rules and an MRP database.

In order to overcome some limits of simulation methodology, researchers have started to develop hybrid approaches integrating other techniques such as fuzzy logic (Brennan & Forough, 1999; Labib et al., 1998; Kazerooni et al., 1997), experimental design (Blosch & Antony, 1999; Li et al., 1997), neural networks (Ito, 2000; Pendharkar 1999) or genetic algorithms (Jahangirian & Conroy, 2000). However, these techniques are practically never used by end users. Simulation packages are under continuous improvement and the future looks promising, but the difficulty of standardising the industrial problems may prevent their mass implementation in industry. Anyway, simulation is a techniques that still has a lot of under-exploited potentialities.

Integrated modelling methods

All the methods and tools described above give specific solutions to several problems in manufacturing systems. However, there is no single conceptual modelling method that can completely model a manufacturing system or most of its sub-systems. As a result of the limitations of methods and techniques, a limited number of integrated modelling methods have been developed:

- GIM Methodology (Doumeingts, Vallespir, & Chen, 1995) integrates three modelling methods: GRAI (to model decisional systems), MERISE (to model

Table 1 Discrete event simulation packages for manufacturing analysis

Simulation software	Class	Vendor-website
Automod	Flow oriented	http://www.autosim.com/
Arena	Flow oriented	http://www.arenasimulation.com/
ModSim	Object oriented	Not supplied anymore
SIMPLE++/eM-Plant	Object oriented	http://www.em-plant.de/
Taylor II	Flow oriented	http://www.showflow.com/
GPSS/H	Object oriented	http://www.wolverinesoftware.com
Simscrip	Flow oriented	http://www.caciasl.com/
Visual simulation	Object oriented	http://www.orcacomputer.com/vse/VSEMain.html
Witness	Flow oriented	http://www.lanner.com/corporate/

information systems) and IDEF0 (to model physical systems). A computerised graphical editor that offers support for the dynamic aspects of manufacturing systems has provided the support for the method.

- SIM (Carrie & Macintosh, 1997) composes two modelling methods: Data Flow Diagrams (DFDs) and GRAI Grids. It is an effective method for modelling manufacturing information systems but it does not consider the dynamic aspects of physical sub-systems in the manufacturing environment.
- GI-SIM (Al-Ahmari & Ridway, 1999) joins three components GRAI grid, IDEF0 and SIMAN in a integrated environment through an interface tool. This tool is used to facilitate user access and data exchange between the activities and functional/simulation models. A global structure of manufacturing systems can be developed using a single level of a modified GRAI grid to illustrate the main functions, decisions and activity centres. The method enhances the static model to include dynamic modelling.
- Kang, Kim, and Park (1998) propose an integrated Modelling Framework for manufacturing systems named IMF-M, that can provide a unified representation of the physical process and the control information system, processes, physical material flows, information objects, resources and materials.
- Zakarian and Kusiak (2001) developed an integrated framework based on IDEF techniques, stream analysis approach and dynamic simulation. This method is oriented to performing the analysis and reengineering of processes and evaluating the impact of changes. The authors propose modifications to the IDEF3 model. Each activity box is divided into two sections with a phrase describing the activity and an expression that describes the mathematical relationship of the output, input and control elements. This representation is used to formulate a dynamic simulation with DYNAMO modelling language representing a set of linked differential

equations describing a closed loop feedback system. This method extends the IDEF3 methodology by including quantitative information. The same authors (Zakarian & Kusiak, 2000) have developed a new analytical approach for process models using IDEF3 objects to create a qualitative and quantitative analysis knowledge base based on the fuzzy Signed Directed Graph (SDG) technique.

Evaluation of techniques

An evaluation of the techniques for manufacturing analysis has been carried out. The review assesses those initiatives attempting to integrate broader ranges of factors that are applicable to a more general extent. The work has been performed through two actions. The first was a systematic review of the capabilities declared by the leading software providers. For this purpose, several modelling tools (BPwin, DesignIDEF, Visio, Grai-Tool) and system simulators (Taylor and Arena) were employed to obtain a general diagnosis for several real manufacturing systems. In all cases analysed, the ultimate goal was to get an overall analysis of a manufacturing system to identify performance indicators and improvement actions with the minimum amount of data. The second action was several interviews with industrial consultants for data acquisition about real experiences for the improvement of manufacturing systems in manufacturing companies from a wide range of sectors (furniture, aerospace, machining, food processing) and sizes, from multinational companies to SMEs. A review of existing literature, mainly benchmark studies and real case studies, was performed during the evaluation process. This evaluation has enabled us to identify those problems yet to be solved and introduce a set of new lines of research. The final conclusions of the study of the existing methods and tools allow us to identify the following weak points:

- (a) *Difficulty in standardising the general analysis of manufacturing systems to seek continuous improvement.* The diversity of manufacturing systems and the variability of the objectives of a general analysis makes it difficult to create a standard method. The reason for this lies in the complexity of manufacturing today, where the production processes are as varied as the range of products manufactured. This affirmation is more evident if we analyse the great number of variables that can be considered. The typology of the demand can involve working with stocks or orders. Sometimes, production consists of low volumes of high engineering products with more or fewer variants but other times, products are characterised by high quality, low cost and high volumes. Additionally the workforce may include a large number of both qualified and unqualified personnel, who may be flexible or rigid. On the other hand, formal and informal systems, good and bad traditions, and new and old cultures are factors that make the manufacturing analysis different for each enterprise. All these factors create a number of influencing factors that complicate the definition of standard methods. Table 2 shows a compilation of the key factors involved, when it comes to defining the scope, objectives and action to be taken in a process designed to analyse manufacturing systems.
- (b) *Methods oriented towards defining strategic decisions.* In most cases, the initiatives related to diagnostic methodologies or the development of indicator systems are focused on strategic aspects of the company. They provide reference frameworks with conceptual value, but they are not sufficiently developed to be applied at the operational level of manufacturing. BPR is a powerful tool for knowledge management but there is a gap with the final process for making tactical decisions in a manufacturing environment.
- (c) *Analysis only descriptive.* Frequently, the results of a manufacturing analysis become simply an audit of the current state of the activities and resources involved in the system. In most cases, the uses of modelling technologies such as IDEF have an unquestionable value of the “AS-IS” state. They make the acquisition of knowledge easier but do not allow orientations about actions to be taken. On many occasions, new TO-BE models cannot be put into practice due to the implementation cost or human problems that have not been reflected in the model. The results only become qualitative models of the activities. Busby (1992) has affirmed that, without the association of quantitative data on production variables (cycle time, cost, efficiency, quality. . .) to the activity models, it proves impossible to identify, order and calibrate the magnitude of the problems posed by the activities. Frequently, methods like IDEF0 only display the identification of where an activity produces an output for another activity and where it does not. In these cases, the model does not provide indicators on production rates or the effects on the system when changing the basic parameters such as time or cost.
- (d) *Absence of interfaces between modelling and analysis tools.* Most of the techniques and methodologies use a graphical representation of objects such as activities and flows to model the functionality of the system. However, these systems are not linked directly with analysis tools such as a dynamic simulator or decision support systems (DSS). Model diagrams cannot be easily converted into a format that can be interpreted by these tools. By analysing the IDEF methods in-depth, we can identify the following problems:
- Some IDEF tools offer interfaces for mapping data into simulation models but interfaces do not support hierarchies. For example, only a single IDEF diagram can be translated into a single simulation model.
 - Not all objects in an IDEF diagram will necessarily map to a simulation model.
 - Not all simulation objects can be modelled into an IDEF model.
 - Difficulties for model quantitative data.
- (e) *Difficulty in quantifying the added value of processes.* Added-value techniques are highly developed in the field of product development, but not in manufacturing engineering. The need to eliminate wastes with no added value in manufacturing systems (transport, inventory, waiting, over production, defects. . .) are concepts that are fully integrated within the improvement strategies of present-day companies, but putting them into practice presents many difficulties. Currently, there are not fully-integrated methods and tools for their universal application.
- (f) *Difficulty in quantifying non-quality costs.* Companies work with good formulations for manufacturing costs, but it is very difficult to find methods that exactly identify the cost in relation to quality wastes (reprocessing, elimination of defective products, inspections). In this sense, ABC/ABM cost techniques applied to manufacturing systems

Table 2 Critical factors in manufacturing systems analysis

Class of activities	Manufacturing system	Products	Industrial sector	Analysis area	Analysis objectives
Set-up	Linear flow	Number of products	Enterprise size	Layout	Manufacturing cost
Operation	Functional	Variants	Competitors	Product in process	Cycle time
Transport	Continuous	Level of technology	Value-added product	Automation	Productivity
Storage	Cellular	Market demand		Maintenance	Labour
Inspection	Fixed position	Level of quality		Information system	Flexibility
Reprocessing				Work method new technologies	

(Gupta, Stahl, & Whinston, 1997) can provide precise knowledge of the all-manufacturing costs.

- (g) *Absence of methods for selecting techniques and planning improvement actions.* The results from a global analysis or from a diagnosis of a manufacturing system can sometimes provide a large number of possible improvement actions. Analysts are faced with a considerable number of manufacturing strategies and technologies, each one claiming to be the best way to improve productivity and reduce manufacturing costs. The results of applying the techniques described in the above sections do not model key factors to study the cost of improvement actions or the availability of resources for implementation. The diversity of techniques makes new prioritisation mechanisms necessary to plan their implementation.
- (h) *Scant real use of modelling techniques in SMEs.* Small and medium size manufacturing enterprises still use classic tools provided by method engineering. Vernadat (2002) affirms that enterprise modelling is almost totally ignored by SME and there is still a long way to go before SME's master these techniques on their own. In our opinion, system-modelling techniques are confined to technological or multinational companies and consultants. The reasons for the their moderate use may be the high cost involved, the time-consuming projects, their complexity, and the profound reengineering knowledge and specialised expertise required for an effective application.
- (i) *Scant real use of advanced techniques.* Simulation is applied to resolve concrete problems (production planning, bottlenecks, scheduling. . .) but not for the overall analysis of a manufacturing system. Advanced tools such as expert systems, fuzzy sets or neural networks are still not used extensively in the manufacturing sector.

New general framework for manufacturing process analysis

New information systems technologies open the way to using different methods and tools in an integrated way. In order to resolve the problems described, a general framework is proposed with the goal of its being used for general manufacturing process analysis (Fig. 3). The development of an integrated modelling solution is currently under development and will be presented in future papers.

The architecture is based on an integrated flow of information from the first phase of capturing data to the decision support process. The components of this architecture are the following.

Data to model

A manufacturing analysis system needs an analytical revision of those data that allow to identify and evaluate the possible specific improvement actions. This goal involves gathering quantitative and qualitative information from the shop floor functional structure (hierarchical structure) right down to the basic production activities (operations, inventory, transport, waiting and inspection. These means use a lot of quantitative data, mainly those associated to each manufacturing activity (cycle times, performance rates. . .) and flow of materials and cost.

Modelling

Modelling tools must be used to represent quantitative and qualitative information. In the case of IDEF models, different attributes could be associated to their objects (activity, input, output, mechanism and control). At present, commercial tools such as All Fusion

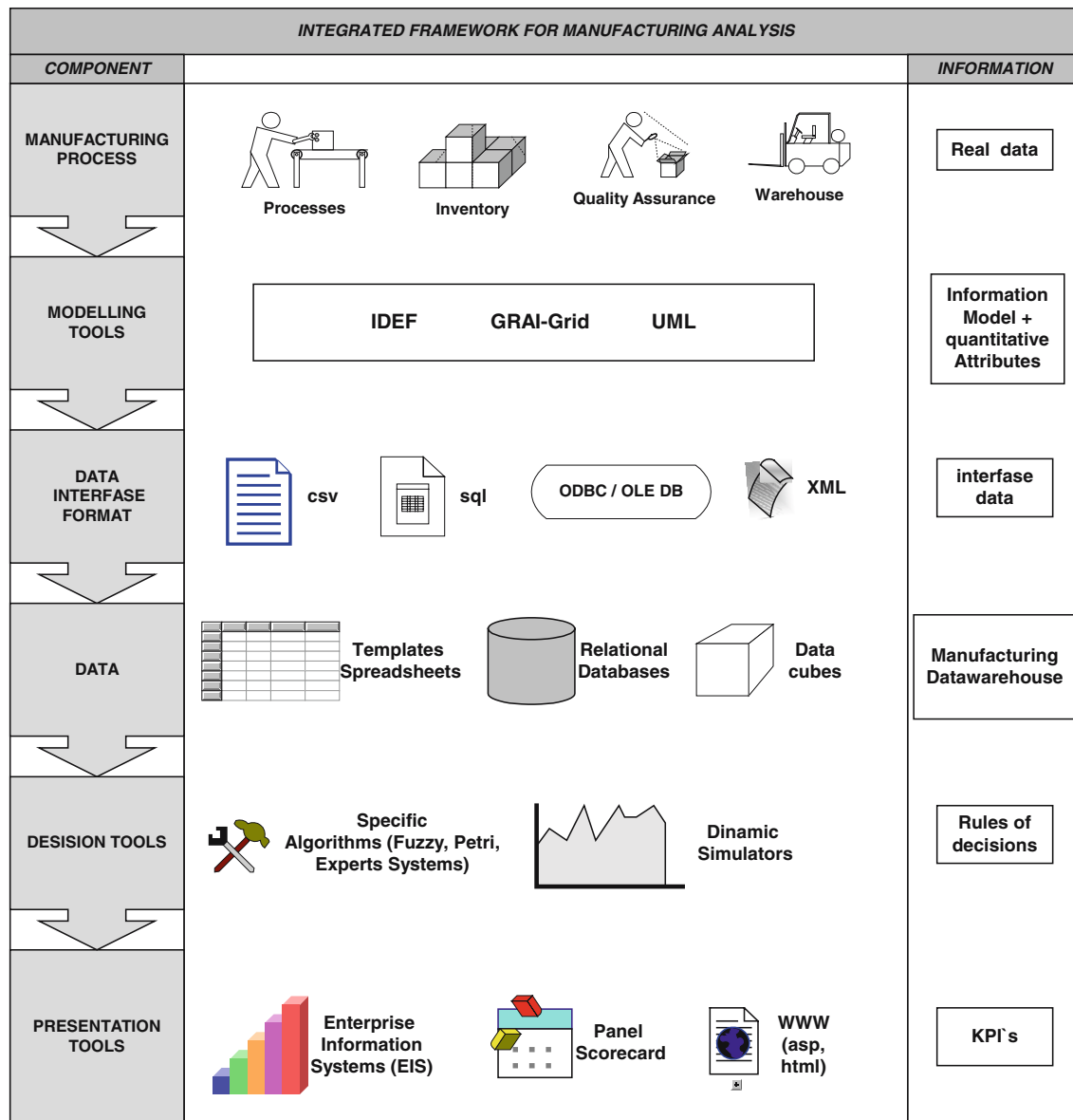


Fig. 3 Framework for manufacturing analysis

Process Modeller (BPWin) allow associating any data (text, numerical, value list) to the IDEF objects throughout UPDs (user defined properties) and attribute libraries. This capability opens the way to associating quantitative or qualitative information to the objects by way of a series of libraries of attributes that are associated to the modelled objects. Each activity type could be associated a set of specific attributes according to the specific requirement system. The attributes must allow subsequent calculation of the main performance indicators of the production system, such as demand by product type, demand by machine, cycle times, quality rates, ABC costs, etc. Normally, this capability has been

used to assign ABC cost to activities but not to incorporate quantitative data related to production variables (cycle time, efficiency, human resources...). In a similar way, the customisation of capabilities of Microsoft VISIO with Visual Basic for applications and Object Model Methods is an alternative solution. It allows the designing of graphic objects with a set of free attributes that could be connected to any external database using ODBC. Recently, these capabilities have used by the manufacturer of Witness Simulator to announce the development of VISIO-templates. Users can use the templates to write directly data to be used later inside the simulation tool. The link between both applications,

VISIO and Witness, is done via XML Files. The incorporation of new attributes and the definition of templates could be considered as an improvement and a modification of the current standard IDEF methods.

Data interface

After creating the quantitative and qualitative data model, the export of this information must be done using structured and standard formats. Modelling tools must be open to simulation analysis and decision support. In this sense, there are several possible solutions:

- Flat text files to be used by preconfigured spreadsheets for each type of manufacturing analysis.
- SQL code to export the attributes as database registers. By generating SQL code from the IDEF models the automatic creation of manufacturing databases is possible without the need to use an IDEF1X model.
- XML files to manage information with web tools.
- Internationally accepted standards for exchange of information. In this point, ISO 10303 STEP (product oriented) and ISO 18629 PSL (process oriented) should be considered.

Manufacturing datawarehouse

Data interface allows creation of data repository for storing all the data necessary to a later evaluation of the system. However, there are always some data that are not linked with modelled activities but are necessary for the analysis. That is the case of data such as the production demand per product, rejected product factor, scrap factors, hourly costs (labour, machine, tools) or data about layout surfaces (machines, warehouse. . .). A solution is the development of interfaces between the manufacturing datawarehouse and the manufacturing database of the ERP systems. Information about product structure, manufacturing routes and MIX production are other examples of valuable data to be used in manufacturing systems analysis.

Decision tools

There are several mechanisms that can be used for evaluating the data:

- The most important is the development of algorithms and equations to transform the information on database into Key Performance Indicators (KPIs) for the continuous improvement process for example. They are used to measure performance and also guide behaviours towards the desired goal, for

example, bottlenecks, utilisation of resource capacity, and general wastes. They are the best way for implementing manufacturing improvements.

- The dynamic analysis of a model may determine the behaviour of the system under a different initial production plan or operational condition in order to perform a repeated experimentation with the model.
- The development of a knowledge-based system to evaluate the state of a manufacturing sub-system and to integrate the decision rules for automatic selection of the best improvement techniques (TPM, SMED, JIT, QFD, . . .) that could be applied in the sub-system. The use of expert systems or fuzzy logic to model the decision process may be a solution for incorporating these rules.
- New techniques of business intelligence (OLAP, cube-data).

Presentation of results

The proposed architecture must be integrated in a scalable workbench where modelling, database transactions, simulation and optimisation techniques allow the information to be easily exploited and the system to be diagnosed. In this sense, the development of metric dashboards is very useful for the deployment of KPIs. The utilisation of Enterprise Information Systems (EIS) gives easy access to the development of specific tools for each application. To quote a particular case, the Forest & Trees software is a valid solution to develop this module in an integrated modelling solution.

Conclusions

The paper presents a detailed, systematic and multidisciplinary evaluation of the methods, techniques and tools that have been used in recent years for analysing manufacturing systems. Some of the main problems identified during the evaluation are the following:

- Difficulty in standardising the general analysis of manufacturing systems.
- Result of analysis too descriptive.
- Difficulty in quantifying the added value of processes and non-quality costs.
- No methods to select techniques and plan improvement actions.
- Problem to incorporate quantitative tools with modelling tools and absence of interfaces with others tools.
- Reduced use of modelling techniques in SMEs.

The results claim the need for new tools that attempt to combine the new system techniques with industrial requirements to eliminate the identified limitations. Manufacturing analysis needs a formulation to transform an information model into high value information, mainly represented through KPIs on metric dashboards. The technical solutions to implement these solutions must be related to: incorporating quantitative and qualitative data to modelling tools, the use of standard connections or standard formats (ODBC, XML, CSV) for creating specific manufacturing databases, the use of new formulations for KPIs and the application of new business intelligence techniques (EIS, OLAP).

As a result of the evaluation, general guidelines for an integrated framework method to integrate manufacturing analysis are proposed. It can be used by end users as a new approach for making better use of modelling techniques for manufacturing process analysis. In the same way, modelling tools developers could use the proposed architecture as a guideline to define future capabilities in new tools.

The work presented is part of a project currently under development in the Manufacturing Department of the Polytechnic University of Madrid with the proposal of developing an integrated modelling framework and tools for the analysis of manufacturing systems. This tool will include: the definition of quantitative–qualitative information in an IDEF3 model, the transformation of the model into manufacturing database data, the development of formulations for calculating KPIs and the design of metric dashboards to diagnose the system identifying and controlling the continuous improvement actions.

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